

1.

```
#include<stdlib.h>
#include<stdio.h>
#include<string.h>
int main()
{
    char *p; int i;
    p=(int*)calloc(12*sizeof(char));
    strcpy(p,"_Sunbeam");
    for(i=1 ; i < 12 ;i++)
    {
        printf("%c",*(p+i));
    }
    free(p); p=NULL;
    return 0;
}
```

- A. _Sunbeam
- B. Sunbeam0000
- C. Sunbeam
- D. Error

Answer:D

2.

What happens if malloc() fails to allocate memory?

- A. It gives Error
- B. It returns a pointer to the last allocated memory block.
- C. It returns the size of the requested memory block.
- D. It terminates the program.
- E. It returns a null pointer.

Answer:E

3.

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int *arr[3];
    arr[0] = (int *)malloc(sizeof(int)*3);
    arr[1] = (int *)malloc(sizeof(int)*1);
    arr[2] = (int *)malloc(sizeof(int)*2);
    for(int i = 0 ; i < 3 ; i++)
    {
        printf(" %u ", arr[i]) ;
    }
    free(arr);
    return 0;
}
```

- A. Print Address of heap section
- B. Print Garbage Values
- C. Memory Leakage
- D. Dangling Pointer

Answer:C

4.

What is the maximum number of bytes that can be allocated with malloc in C?

- A. It depends on the system and available memory
- B. 2^{16} bytes
- C. 2^{32} bytes
- D. 2^{64} bytes

Answer : A

5.

```
#include<stdio.h>
#include <stdlib.h>
#include <string.h>
int main()
{
    char *p=NULL;

    p = calloc(sizeof(char) * 15, 1);
    strcpy(p, "#Sunbeam");
    printf("%s ", p + ((p + 3) - (p + 2)));
    strcpy(p, "Sunbeam-Karad");

    int x = ((p + 6) - (p + 3));
    p = realloc(p, sizeof(char) * ((p + 9) - (p + 4)));
    printf("%s ", p + x);
    free(p);

    return 0;
}
```

- A. Sunbeam nbeam-Karad
- B. Sunbeam nbeam- K
- C. Sunbeam Sunbeam-Karad
- D. Sunbeam beam-Karad

Answer: D